

SUBJECTIVE WORKSHEET

Life Processes in Plants

Science — Biology · Class 7 · Max Marks: 34 · Time: 1 hour

Section A — Very Short Answer ($1 \times 5 = 5$)

1. Name the green pigment that traps sunlight in a leaf. [1]

2. Name the gas released during photosynthesis. [1]

3. Name the tissue that carries water in a plant. [1]

4. Name one insectivorous plant. [1]

5. What is the loss of water vapour from leaves called? [1]

Section B — Short Answer ($2 \times 5 = 10$)

1. Write the word equation for photosynthesis. [2]

2. Why are green plants called autotrophs? [2]

3. Give two differences between xylem and phloem. [2]

4. Why do insectivorous plants trap insects even though they are green? [2]

5. State two ways in which transpiration is useful to a plant. [2]

Section C — Long Answer ($5 \times 3 = 15$)

1. Explain the process of photosynthesis, stating the raw materials, the conditions needed and the products formed. [5]

2. Describe the modes of heterotrophic nutrition in plants (parasitic, insectivorous, saprotrophic, symbiotic) with one example of each. [5]

3. Explain how water absorbed by the roots reaches the leaves at the top of a tall tree. [5]

Section D — Higher-Order Thinking (Give Reasons) ($2 \times 2 = 4$)

1. A potted plant is kept in a completely dark room for a week but watered daily. Will it stay healthy? Give a reason. [2]

2. A cut stem is placed in red-coloured water. Explain why red streaks appear moving upward but not downward. [2]

Answer Key

Very Short Answer

1. Chlorophyll.
2. Oxygen.
3. Xylem.
4. Pitcher plant (or Venus flytrap).
5. Transpiration.

Short Answer

1. Carbon dioxide + water $\xrightarrow{\text{sunlight, chlorophyll}}$ glucose (food) + oxygen.
2. Because they make their own food from simple substances by photosynthesis, instead of taking ready-made food.
3. Xylem carries water and minerals upward from the roots; phloem carries the food made in the leaves to all parts of the plant.
4. They grow in soil poor in nitrogen; they make their own food by photosynthesis but trap insects to get the nitrogen they need.
5. It helps pull water and minerals up from the roots, and it cools the plant.

Long Answer

1. In photosynthesis, green leaves use carbon dioxide (from air, via stomata) and water (from the soil, via roots and xylem) in the presence of sunlight and chlorophyll to make glucose (food) and release oxygen.
2. Parasitic — takes food from a living host (Cuscuta). Insectivorous — traps insects for nitrogen (pitcher plant). Saprotrophic — feeds on dead/decaying matter (mushroom/fungi). Symbiotic — two organisms live together for mutual benefit (lichen: alga + fungus).
3. Root hairs absorb water and minerals from the soil; these enter the xylem, which forms continuous tubes up the stem. As water evaporates from the leaves (transpiration), the pull created draws the water column up the xylem to the leaves.

Higher-Order Thinking (Give Reasons)

1. No. Without light it cannot photosynthesise, so it makes no food and becomes weak (and may die).
2. The xylem carries water upward from the roots to the leaves, so the coloured water rises through the xylem and does not go downward.